

Instructions:

- **Due February 25, 11:59PM uploaded to Blackboard as a single pdf.**
- Each problem is worth $\frac{100}{\text{number of problems}}$ points.
- Start each problem on a new page and put a box around your final answer to help the grader.
- Label your file as `ENME403_S26_HW04_yourname.pdf`.
- If you use Simulink to generate results for the homework submission, include screenshots of the block diagrams and all relevant plots.
- **Use of AI Tools:** The use of AI software to prepare your submission is strictly prohibited. If we determine that AI generated a submission, the work will receive a grade of zero and will be reported to the Academic Conduct Committee. You may use AI tools to aid your understanding, but all submitted work must be entirely your own.
- Failure to follow these guidelines will result in zero points.

Learning Objective: This homework reinforces the ability to compute and apply Laplace transforms, derive transfer functions from physical system models, use the Final Value Theorem to predict steady-state behavior, and connect state-space and transfer-function representations of linear systems.

Problem 1. Write the Laplace transform of the function $f : \mathbb{R} \rightarrow \mathbb{R}$, where, for all $t < 0$, $f(t) = 0$, and for all $t \geq 0$,

1. $f(t) = e^{at}$
2. $f(t) = \sin(\omega t)$
3. $f(t) = e^{at} \sin(\omega t)$

You do **not** need to derive the transforms; simply state the final expressions of the corresponding Laplace transform.

Problem 2. Let $F(s) = \mathcal{L}[f(t)]$.

1. Using $\mathcal{L}\left[\frac{d}{dt}f(t)\right] = sF(s) - f(0)$, calculate $\mathcal{L}[\cos(\omega t)]$.
2. Using $\mathcal{L}[e^{at}f(t)] = F(s - a)$, calculate $\mathcal{L}[e^{at} \sin(\omega t)]$.
3. Using $\mathcal{L}[f(t - \tau)] = e^{-s\tau}F(s)$, calculate $\mathcal{L}[\delta(t - t_0)]$.
4. Using $\mathcal{L}[tf(t)] = -\frac{d}{ds}F(s)$, calculate $\mathcal{L}[t]$.

Problem 3. Write the transfer function from the input f to the position q for the following systems.

1. URB
2. DRB
3. UO
4. DO

If applicable, use the final value theorem (FVT) to predict the asymptotic value of the position, in terms of m, c, k , for each of the systems above for i) an impulse input and ii) a unit-step input. Assume zero initial conditions. If the FVT is not applicable, explain why.

Problem 4. Consider the DO

$$m\ddot{q} + c\dot{q} + kq = f. \quad (1)$$

The DO in the state-space form is

$$\dot{x} = Ax + Bu, \quad (2)$$

$$y = Cx + Du, \quad (3)$$

where

$$A \triangleq \begin{bmatrix} 0 & 1 \\ -k/m & -c/m \end{bmatrix}, \quad B \triangleq \begin{bmatrix} 0 \\ 1/m \end{bmatrix}, \quad C \triangleq [1 \quad 0], \quad D \triangleq 0. \quad (4)$$

Compute $C(sI - A)^{-1}B$ by hand. What is this expression equal to?

Hint: For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

Extra Credits Problem 1. [25 points.] Let $F(s) = \mathcal{L}[f(t)]$. Show that

1. $\mathcal{L}\left[\frac{d}{dt}f(t)\right] = sF(s) - f(0)$. [Hint: Use integration by parts shown below.]
2. $\mathcal{L}[e^{at}f(t)] = F(s - a)$.
3. $\mathcal{L}[f(t - \tau)] = e^{-s\tau}F(s)$.
4. $\mathcal{L}[tf(t)] = -\frac{d}{ds}F(s)$.
5. $\mathcal{L}\left[\frac{f(t)}{t}\right] = \int_s^\infty F(s)ds$.

Integration by Parts

The integration by parts formula is

$$\int_a^b f(t) \frac{d}{dt}g(t)dt = f(b)g(b) - f(a)g(a) - \int_a^b g(t) \frac{d}{dt}f(t)dt. \quad (5)$$

Let $f(t) = e^{-st}$. Then,

$$\int_a^b e^{-st} \frac{d}{dt}g(t)dt = e^{-sb}g(b) - e^{-sa}g(a) + s \int_a^b g(t)e^{-st}dt. \quad (6)$$